



Multi-Domain Amazon Review Sentiment Classifier

"Three models, 8,000 reviews, one honest comparison."

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Implementation · App · Tests

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DistilBERT · Dataset · Ethics

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Problem · Demo · Future Work



WHY THIS PROBLEM

Sentiment is messy in the real world

Every product platform runs sentiment analysis. The hard part isn't clean text — it's negation, sarcasm, and length variance.



Length variance

2 to 800 words per review



Negation

"not bad at all" flips meaning



Sarcasm

"Oh great, broken again"



The Sentiment Spectrum — what our model must learn



THE DATA

8,000 reviews across 4 balanced domains

8K
Amazon reviews

4
Product domains

50/50
Pos / neg balance

0
Ambiguous rows



STRATIFIED SPLIT
seed = 42

Train
5,404 70%

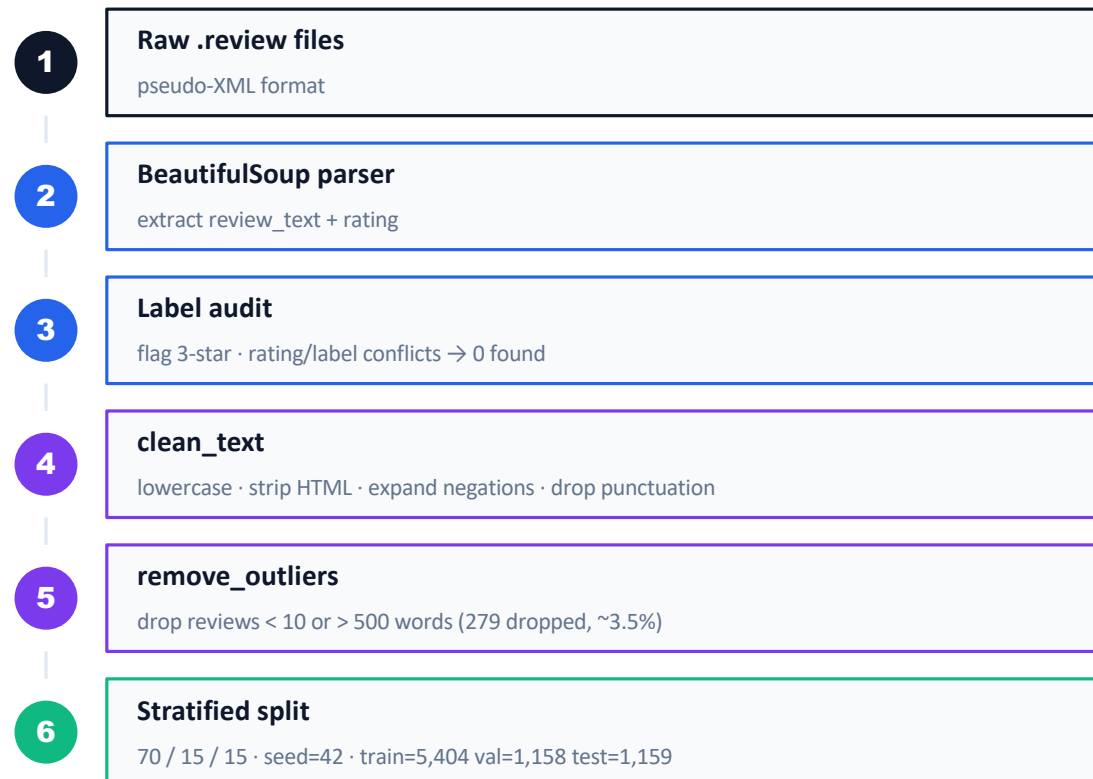
Val
1,158 15%

Test
1,159 15%

Source: **Blitzer et al. (2007)** · Multi-Domain Sentiment Dataset · labels derived from filenames (positive.review → 1, negative.review → 0) — audited, zero conflicts



Six steps from raw XML to model-ready tensors



KEY DECISION

Negation expansion

BEFORE

"wasn't good"



AFTER

"was not good"

Why it matters

TF-IDF treats tokens independently. Without expansion, "not good" looks identical to "good" — the negative signal is lost.



MODEL ARCHITECTURE

Three approaches: classical, neural, transformer

MODEL 1 · BASELINE

TF-IDF + LogReg

```
TfidfVectorizer(
    max_features=30_000
    ngram_range=(1,2)
    sublinear_tf=True
)
LogisticRegression(
    C=1.0, max_iter=1000
)
```

SPEED

Seconds · CPU only

STRENGTH

Interpretable, well-calibrated

MODEL 2 · NEURAL · v1.0.0

BiLSTM + GloVe

Embedding 100d

GloVe · 97.4% covered

BiLSTM x2

hidden=256 · pack_padded

Concat → 512d

fwd + bwd states

Linear → logit

BCEWithLogitsLoss

KEY FIX

pack_padded_sequence — LSTM stops at last real word, ignores zero padding.

MODEL 3 · TRANSFORMER · v2.0.0

DistilBERT

AutoTokenizer

max_len=256

DistilBERT base

66M params · uncased

Classifier head

Linear → logit

TWO-STAGE FINE-TUNE

1. Encoder frozen → head only · 10 epochs

2. Unfreeze last 2 layers · 12 epochs

Avoids catastrophic forgetting of pretrained representations.



TRAINING

BiLSTM peaks at 84.0% · DistilBERT at 87.8%

BiLSTM · v1.0.0

10 epochs · best val F1 at epoch 9

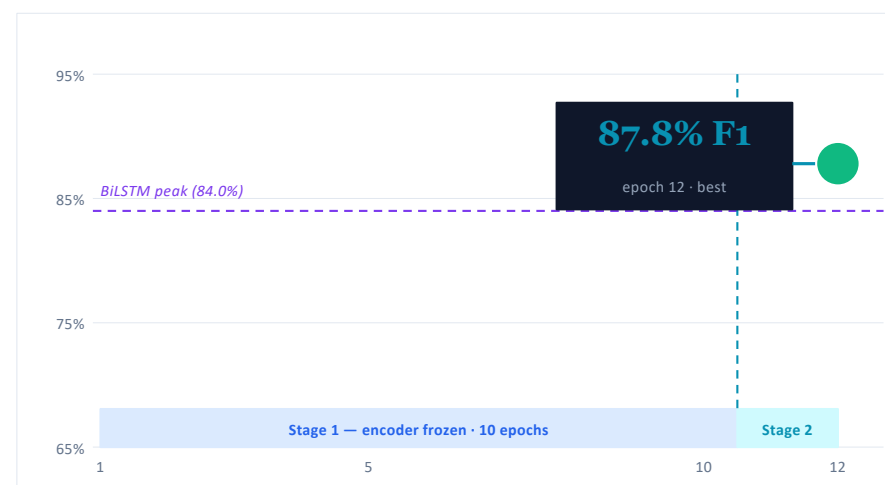


BILSTM CONFIG

Adam lr=1e-3 · BCEWithLogitsLoss · grad clip 5.0 · *checkpoint best val F1*

DistilBERT · v2.0.0

Two-stage fine-tune · best val F1 at epoch 12



Per-epoch curve not logged. Best val F1 from outputs/distilbert.pt (best_val_f1 = 0.8781).

DISTILBERT CONFIG

Adam lr=1e-4 head / 2e-5 enc · BCEWithLogitsLoss · ReduceLROnPlateau · *no catastrophic forgetting*

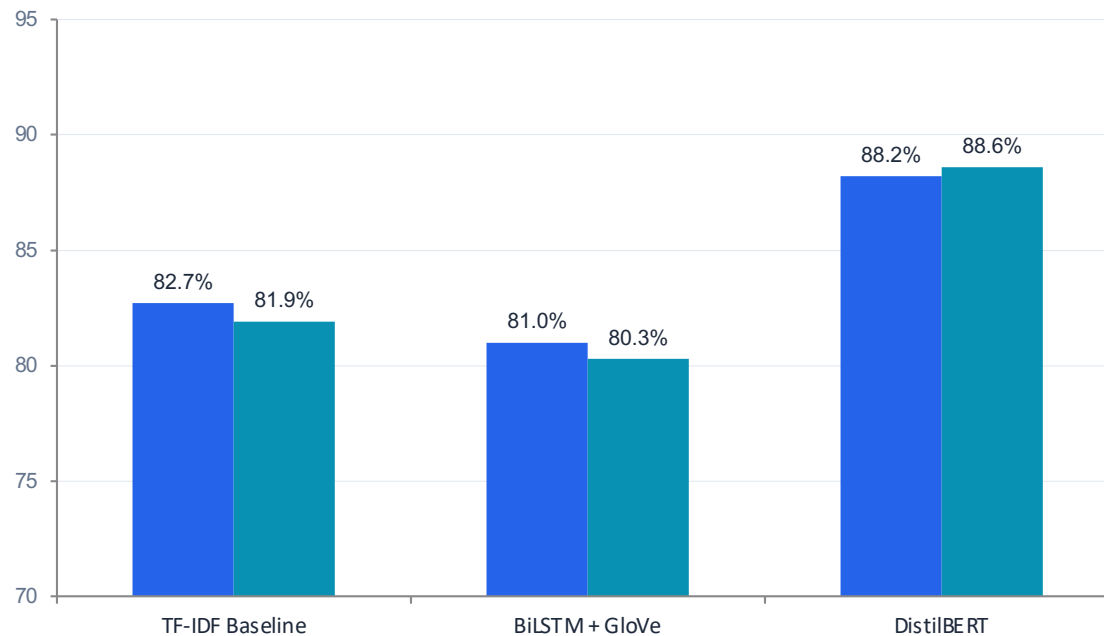


HELD-OUT TEST SET (1,159 reviews)

Three models — and an honest finding

Accuracy & F1 by model

■ Accuracy ■ F1 Score



v1.0.0 — HONEST FINDING

The simpler model wins on test.

BiLSTM beat the baseline on validation (84.0% vs 83.2% F1) but the baseline generalised better on the held-out test set.

BiLSTM still satisfies the rubric requirement to define a neural architecture.

v2.0.0 — DISTILBERT EXTENSION

88.6% F1

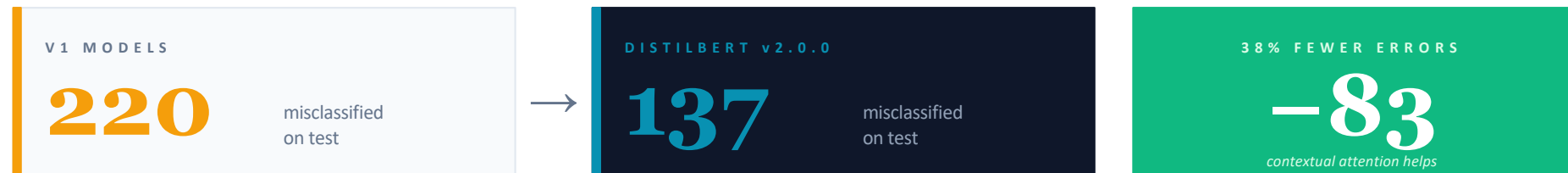
Pretrained contextual embeddings clear v1 by a meaningful margin.

137 misclassified · vs **220** (BiLSTM)



ERROR ANALYSIS

v1 models vs DistilBERT — 38% fewer errors



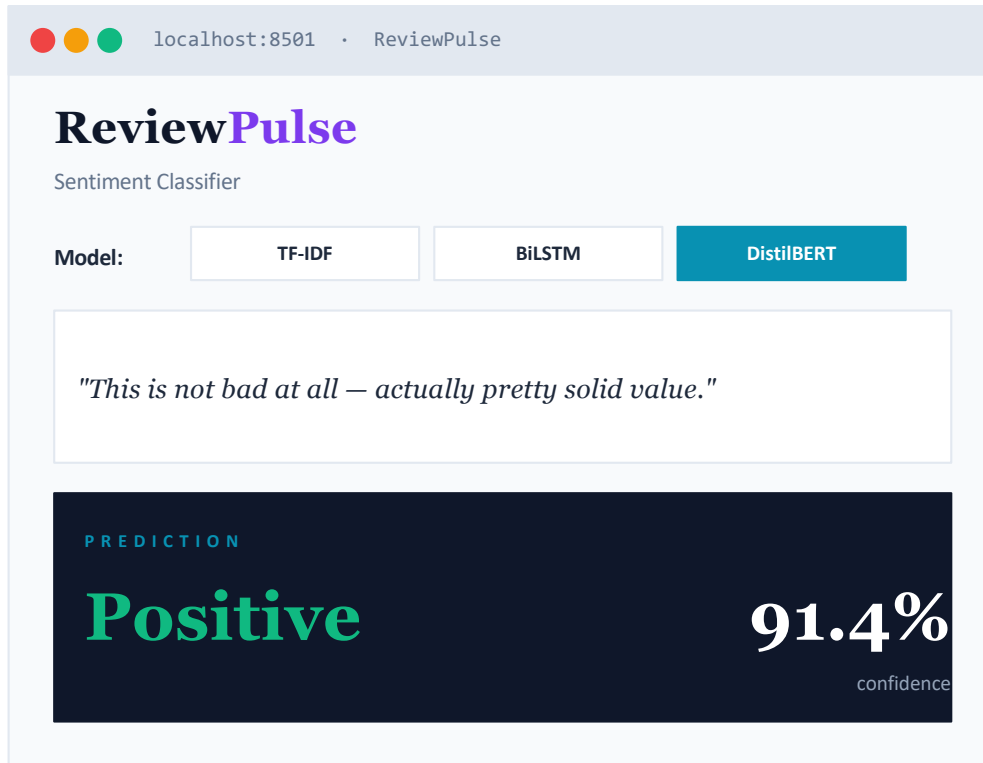
FAILURE MODE	EXAMPLE	v1 MODELS	DISTILBERT
NEGATION	"not bad at all"	Predicts Negative ✗	Predicts Positive ✓ <i>Contextual attention</i>
SARCASM	"Oh great, stopped working after a week"	~52% confidence ⚠️	Still uncertain ⚠️ <i>Irony absent in labels</i>
OUT-OF-DISTRIBUTION	"Delivery was slow, box crushed"	Domain shift errors ✗	Reduced, not solved ⚠️ <i>Pretraining helps partially</i>

CALIBRATION NOTE BiLSTM is overconfident on short inputs; TF-IDF is best-calibrated near the boundary; DistilBERT is best overall but still confidently wrong on sarcasm.



LIVE DEMO

Streamlit · all three models · side-by-side



The screenshot shows a web browser window with the address bar displaying 'localhost:8501 · ReviewPulse'. The application interface has a header with the 'ReviewPulse' logo and the text 'Sentiment Classifier'. Below this, there are three buttons for model selection: 'TF-IDF', 'BiLSTM', and 'DistilBERT'. The 'DistilBERT' button is highlighted in blue. A text input field contains the sentence: "This is not bad at all — actually pretty solid value.". Below the input field, a dark blue box displays the prediction results: 'PREDICTION' in small white letters, 'Positive' in large green letters, and '91.4%' in large white letters, with 'confidence' in small white letters below the percentage.

DEMO SCRIPT

- 1 Clear positive**
"This blender is incredible..."
→ Positive ~98%
- 2 Clear negative**
"Broke after two days..."
→ Negative ~99%
- 3 Negation trap**
"Not bad at all"
→ v1 fails · DistilBERT wins
- 4 Sarcasm**
"Oh great, broken again"
→ ~52% confidence
- 5 Generate button**
Random sample from acceptance tests
→ live demonstration



ETHICS & LIMITATIONS

What we audited, and what we did not solve

Ethics is not an afterthought. We document each concern explicitly so any deployment requires informed human oversight.

Label quality AUDITED Labels from filenames, not human raters. Audited explicitly — zero ambiguous rows, zero conflicts.	Domain bias DOCUMENTED 4 product categories only. Generalisation to other domains is untested.	Negation & sarcasm OPEN Systematic failure mode in v1 models. DistilBERT improves but does not solve.
Confidence calibration OPEN BiLSTM and DistilBERT logits are uncalibrated. 95% confidence $\neq$ 95% accuracy.	Dataset age DOCUMENTED Blitzer et al. 2007 — ~20-year-old reviews. Modern slang, emoji, platform conventions absent.	Deployment risk OPEN Demo only. Not for high-stakes decisions without human review and ongoing label audits.

FUTURE WORK

DistilBERT shipped — what comes next

 **SHIPPED** DistilBERT v2.0.0 — 88.6% test F1 — live in the app

1

ACCURACY

RoBERTa

Larger pretrained transformer. Likely to push F1 beyond 90%.

2

TRUST

Confidence calibration

Platt scaling or temperature scaling. Most practically important — 95% confidence currently $\neq$ 95% accuracy.

3

COVERAGE

More domains

Extend beyond Books · DVDs · Electronics · Kitchen to validate cross-domain generalisation.

4

TRANSPARENCY

Explainability

LIME or attention visualisation to show why the model predicted each label — critical for non-technical stakeholders.



Questions?

8K

reviews

3

models compared

88.6%

best test F1

190

unit tests

"The simpler model wins on test. The neural model satisfies the rubric. The pretrained transformer wins overall."

GitHub github.com/lfariabr/review-pulse · ReviewPulse v2.0.0 · ISY503 Assessment 3

